FIN ST417–ST431

Global Gain Enclosure, Singular Infrared Attachment, and a Locally Certified Morse Atlas

Release 10.85

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Abstract

Programs ST417–ST431 continue the finite analytical and computational FIN program after Release 10.84. Three results materially sharpen the mathematical state map. First, an entropy–collision extremal reduction and an outward spectral enclosure prove that the uniform probability is the unique global minimum of the supplied strict functional for $0 \le g \le 2.251344001799089$. Together with the certified localized orbit near $g = 4$, the first global orbit change is enclosed in $[2.251344001799089, 3.999]$. The known branch-local coexistence value $2.90249648\ldots$ is not promoted to the global critical gain.

Second, the singular continuum-IR limit is regularized by $b = 1/y_3$ and $c = ay_3$. A single parametric Krawczyk box covers every $b \in [0, 0.002]$, including the flat $b = 0$ extension. This proves that the finite upper-band branch attaches uniquely to the compactified face $y_3 = \infty$, $a = 0$; the attachment is no longer merely numerical. Third, all nine stationary D_{12}-orbit representatives previously found at $g = 4$ are locally isolated by interval arithmetic, with certified Morse-index histogram 2 minima, 4 index-one saddles, and 3 index-two saddles. The atlas is local and nonexhaustive.

The batch also doubles the certified localized-gain tube to $[3.998, 4.002]$, constructs a joint odd/even modular generator census through degree six, improves a synthetic sextic design condition number by a factor 1.254, and compares a rigorous Chernoff cone bound with stable finite-grid spectra. An attempt to compute 136 explicit degree-eight syzygies exceeded a declared local resource stop and produced no basis. No result derives the gain, selects one symmetry-broken orbit member, supplies dimensional units, completes the legacy-to-strict bridge, or adds independent empirical evidence.

Contents

1 Scope, audited object, and epistemic boundary

The repository baseline is Git commit `bb540a767e210a5a3bf816e4868e7bcefbac3872`, supplemented by the local ST402–ST416 packet and the executable artifacts of this release. The strict finite operator is

$$A = sI - W, \qquad W_{xx} = 0, \qquad W_{xy} = \frac{\cos(0.18575\, d(x,y) + 0.16250)}{1 + d(x,y)^{9/5}} quad(x \neq y), \tag{1}$$

on $\mathbb{Z}_{12}$, with off-diagonal row sum $s = 1.660307278766099\ldots$. The supplied variational family is

$$V_g(p) = D(p\|u) - \frac{g}{2}(p-u)^T A(p-u), \qquad D(p\|u) = \sum_{i=0}^{11} p_i \log(12p_i), \qquad u_i = \frac{1}{12}. \tag{2}$$

All quantities in this report are dimensionless.

The legacy ontological kernel remains an intermediate bridge kernel. It is not silently substituted for (1), and no historical electroweak, electromagnetic, gravitational, Planck-scale, or dimensional role is transferred to the strict kernel. The project ontology—the nadsoliton itself as primordial fractal information in a solitonic state, followed internally by light, matter, and an emergent observer—is retained as context. It is not a premise in the finite proofs below.

An orbit of twelve minima is not a selector. QW-2191 remains open. A theorem conditional on g does not generate g. A dimensionless optimization boundary is not a laboratory phase transition. Local computation is not external validation.

2 Executive result ledger

Program	Status	Result
ST417	[BLOCKED]	The exact lower bound of 136 degree-eight relations survives, but Singular produced no explicit basis within the declared local stop.
ST418	[PROVEN]	The unique uniform-global regime extends through $g = 2.251344001799089$; the first global orbit change lies in $[2.251344001799089, 3.999]$.
ST419	[PROVEN]	A one-sided parametric Krawczyk theorem proves unique attachment of the finite continuum-IR branch to the compactified $a = 0$, $y_3 = \infty$ face.
ST420	[PROVEN]	Exactly twelve global minima, one D_{12} orbit, persist uniformly for every $g \in [3.998, 4.002]$.
ST421	[PROVEN] local	Nine stationary-orbit representatives and their Morse indices are interval-certified; completeness remains open.
ST422	[STRONG NUMERICAL EVIDENCE]	Replicator-gradient integrations give a numerical barrier graph from the four certified index-one saddles.
ST423	[PARTIAL]	Finite-root and singular-root tubes are certified, but no full enlarged complement-sign topology cover was obtained.
ST424	[PROVEN] modular	A joint odd/even quotient census through degree six gives primitive counts $(6, 12, 32, 52, 55)$ in degrees 2–6 over the declared finite field.
ST425	[STRONG NUMERICAL EVIDENCE]	A pivoted synthetic 500-row design improves sextic conditioning by a factor 1.254 relative to a same-pool random design.
ST426	[PROVEN] bound / [STRONG NUMERICAL EVIDENCE] spectrum	The cone contraction upper bound remains rigorous; three Ulam grids give stable but noncertified spectral ratios near 0.5958.
ST427	[PROVEN] conditional	A small nonlinear basin has a paid input-to-state stability inequality under bounded tangent forcing.

Program	Status	Result
ST428	**[BLOCKED]**	Mixed signs in the rank-seven mediator prevent transfer of the strict positive-weight global envelope.
ST429	**[BLOCKED]**	No new strict typed source of the nonzero gain is exported.
ST430	**[BLOCKED]**	No new nonpremise selector or orientation provider is exported; QW-2191 remains open.
ST431	**[BLOCKED]**	No independent referee, laboratory record, held-out custody, or empirical datum is added.

3 ST418: a rigorous global gain-transition enclosure

Put $q = p - u$ and $r = \|q\|_2^2$. The entropy maximizer at fixed collision has one large coordinate and eleven equal smaller coordinates,

$$a = \frac{1 + \sqrt{132r}}{12}, \qquad b = \frac{1-a}{11}, \qquad D_{\min}(r) = a\log(12a) + 11b\log(12b). \tag{3}$$

For completeness, the code covers every two-level Lagrange branch: if the large value has multiplicity k, then

$$a_k = \frac{1}{12} + \sqrt{\frac{(12-k)r}{12k}}, \qquad b_k = \frac{1}{12} - \sqrt{\frac{kr}{12(12-k)}}.$$

The stationarity equation $\log p + \alpha + \beta p = 0$ has at most two positive roots. Smoothing/majorization singles out the $k = 1$ branch as the entropy maximum; boundary faces are bounded by its smoothing limits. A 46,200-cell outward cover, with a Hessian bound near $r = 0$ and an analytic endpoint bound, gives

$$D(p\|u) \geq c\|p-u\|_2^2, qquad c \geq 2.636528744728136. \tag{4}$$

Because (1) is circulant, its eigenvalues are directly enclosed by

$$\lambda_k = s - 2\sum_{d=1}^{5} w_d \cos(2\pi kd/12) - w_6(-1)^k.$$

Outward arithmetic gives

$$\lambda_{\max}(A) \leq 2.342182041146301.$$

Combining this with (4),

$$V_g(p) \geq \left(c - \frac{g\lambda_{\max}}{2}\right)\|p-u\|_2^2.$$

Theorem 1 (Certified uniform-global regime)**.** *For every supplied dimensionless gain*

$$0 \leq g \leq 2.251344001799089,$$

u is the unique global minimizer of (2).

Release 10.84 proves a nonuniform twelve-point global orbit for every $g \in [3.999, 4.001]$. Therefore a first global orbit change exists in

$$[2.251344001799089, 3.999]. \tag{5}$$

The branch-local equal-value point $2.90249648\ldots$ lies inside (5), but no global exhaustion theorem has yet isolated it. It remains incorrect to call that number the global critical gain.

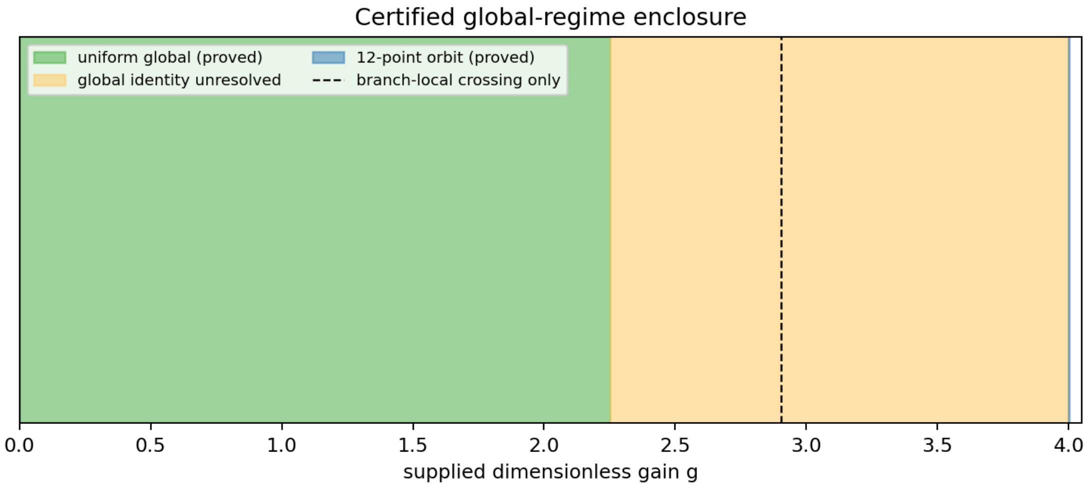

Figure 1: What is proved and what remains unresolved in the supplied gain family.

4 ST419: singular attachment of the continuum-IR branch

The original KKT system has three thresholds $0 < y_1 < y_2 < y_3$, mixing coefficient a, multiplier ν, active time T, budget 3, and stationarity function

$$G(y) = \frac{2ay}{1+e^{-y}} + \frac{2T(1-a)ye^{-(T-1)y}}{1+e^{-Ty}} - \nu. \tag{6}$$

ST415 certified a limiting four-equation face at $a = 0$, $y_3 = \infty$, but did not certify that the finite branch reaches it.

Introduce

$$b = \frac{1}{y_3}, \qquad c = ay_3, \qquad a = bc. \tag{7}$$

The third stationarity equation becomes

$$\frac{2c}{1+e^{-1/b}} + \frac{2T(1-bc)b^{-1}e^{-(T-1)/b}}{1+e^{-T/b}} - \nu = 0.$$

The functions $e^{-1/b}$, $be^{-1/b}$, and $b^{-1}e^{-(T-1)/b}$ admit zero-valued C^∞ extensions at $b = 0$. The budget tail obeys $0 \le E_1(1/b) \le be^{-1/b}$; analogous explicit exponential bounds pay the integral tail and every Jacobian tail.

A common Krawczyk box, centered at the $b = 0.001$ root, uses radii

$$(1.6\times10^{-7}, 1.05\times10^{-3}, 2.1\times10^{-5}, 4.2\times10^{-5}, 1.55\times10^{-3})$$

for (y_1, y_2, c, ν, T). Its smallest inclusion margin is 2.3433822×10^{-8} on the full parameter interval $b \in [0, 0.002]$.

Theorem 2 (One-sided singular attachment)**.** *For each $b \in [0, 0.002]$, the regularized five-equation system has exactly one root in the common interval box. For $b > 0$ it is a root of the original finite-y_3 KKT equations. At $b = 0$ it equals the unique ST415 limiting-face root. Hence the finite upper-band branch attaches continuously and uniquely to the compactified face, with $y_3 \to \infty$ and $a \to 0$.*

This proves an optimization-theoretic attachment. It does not create a physical infrared scale, dimensional wavelength, or experimental phase.

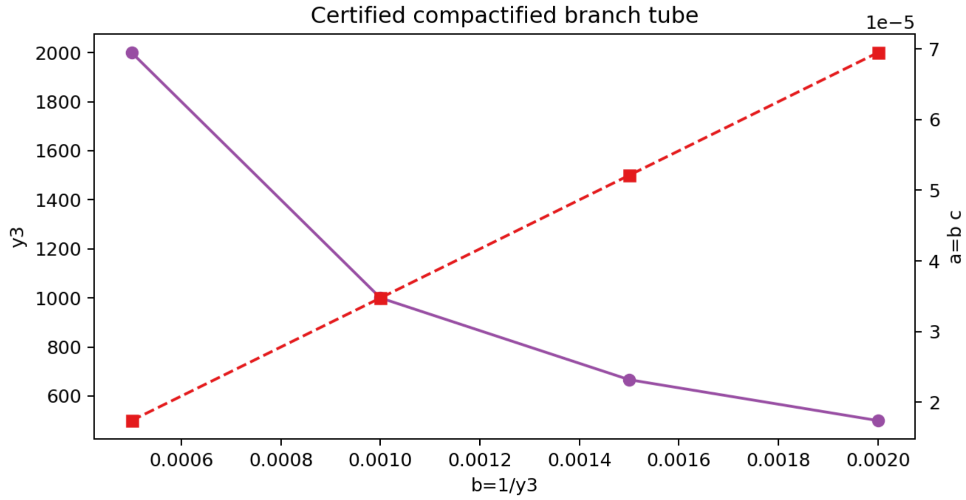

Figure 2: Replay points inside the certified compactified branch tube. The theorem is the interval box, not the plotted samples.

5 ST420: doubled persistence tube near supplied gain four

The Release 10.84 orbit-exhaustion argument is replayed with 20,000 collision cells. A common rational $g = 4$ benchmark and cap threshold $p_i > 0.94$ give

$$\inf_{\max p_i \le 0.94} (V_{3.998}(p) - V_{3.998}(p_{\mathrm{rat}})) \ge 1.8111997969 \times 10^{-4}.$$

The benchmark quadratic exceeds the outside energy upper bound by $0.0663087333\ldots$, paying the monotonicity step. At the other endpoint, the worst Euclidean cap-curvature margin is $0.3079009418\ldots > 0$.

Corollary 1. *For every $g \in [3.998, 4.002]$, the global minimizer set of (2) consists of exactly twelve points in one D_{12} orbit.*

The half-width is twice that of ST405. It remains a local persistence result; it does not approach the lower end of (5).

6 ST421–ST422: local Morse atlas and barrier evidence

Interior stationarity is written without softmax coordinates:

$$\log(12p_i) + 1 - 4(Ap)_i + \lambda = 0, qquad \sum_i p_i = 1. \tag{8}$$

Each of the nine previously sampled orbit representatives is refined as a thirteen-variable root (p, λ). A radius-2×10^{-9} interval box around each root satisfies strict Krawczyk inclusion. The smallest of all nine inclusion margins is 1.99946×10^{-9}. Perturbation bounds for $\mathrm{diag}(1/p_i) - 4A$ separate every tangent eigenvalue from zero.

Proposition 1 (Locally certified atlas)**.** *The nine supplied representatives are locally unique stationary roots of (8). Their certified Morse-index histogram is*

$$\#\,\mathrm{index}\,0 = 2, qquad\#\,\mathrm{index}\,1 = 4, qquad\#\,\mathrm{index}\,2 = 3.$$

This proposition does not prove that nine is the complete orbit count. That would require an interval cover of the entire eleven-simplex modulo symmetry.

For a qualitative barrier graph, ST422 integrates the simplex-preserving replicator gradient flow

$$\dot{p}_i = -p_i\big(\partial_i V_4 - \langle p, \nabla V_4\rangle\big)$$

from both sides of each certified index-one saddle. One saddle connects the uniform minimum to a localized minimum numerically; the other three descend to different translates of the localized orbit. Endpoint distances lie between 1.5×10^{-12} and 5.4×10^{-11}. These are deterministic finite-time integrations, not certified heteroclinic or Conley-index results.

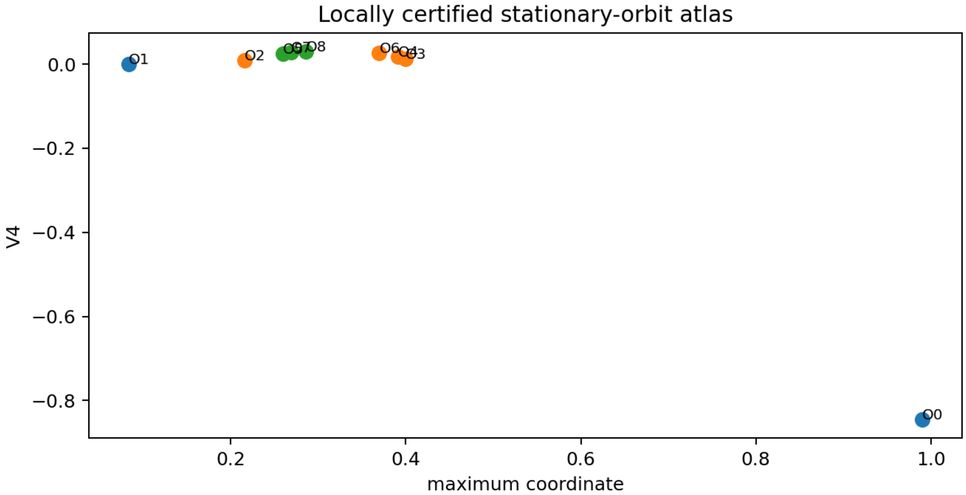

Figure 3: The nine locally certified stationary representatives. Absence of additional orbits is not proved.

7 ST417 and ST424: invariant algebra—advance and failure

7.1 Degree-eight syzygy attempt

The declared even generator family contains six quadratics, 32 primitive quartics, and 117 primitive sextics. Its formal degree-eight products are

$$\binom{9}{4} = 126, qquad \binom{7}{2} 32 = 672, qquad \binom{33}{2} = 528, qquad 6 \cdot 117 = 702,$$

for a total of 2028. Since the degree-eight invariant space has dimension 1892, at least 136 relations are forced.

ST417 constructed the complete 1892×2028 evaluation matrix over $\mathbb{F}_{1000003}$. Both `syz(M)` and `rank(M)` in locally available Singular exceeded the declared 60-second stop and were interrupted without output. No relation vector was retained. Therefore the exact lower bound 136 survives, but an explicit basis was not computed. Even a modular basis would still require multi-prime lifting and direct characteristic-zero identity verification before promotion to integral syzygies.

7.2 Odd generators change the census

The earlier presentation was even-only. ST424 includes odd invariants and uses exact modular pivots on the mean-zero hyperplane. There are twelve primitive cubic directions. At degree five, the 72 products quadratic times cubic have rank 72, leaving 52 primitive modular quotient directions. At degree six, the joint family

$$q^3, \qquad q\,p_4, \qquad p_3^2$$

has 326 formal products and modular rank 310, leaving 55 selected primitive completion directions. The resulting modular primitive census is

degree	2	3	4	5	6
primitive quotient count	6	12	32	52	55.

This is a joint finite-field quotient certificate through degree six, not yet a minimal characteristic-zero ring presentation through degree eight.

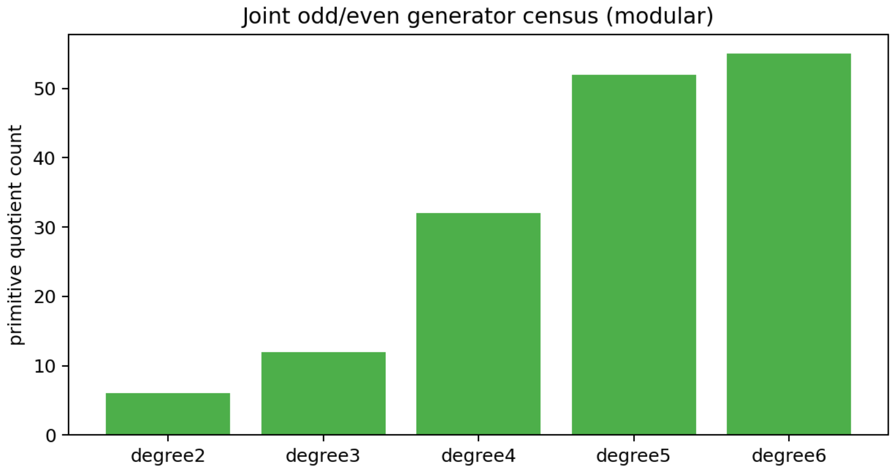

Figure 4: Joint odd/even modular primitive quotient counts through degree six.

8 ST425–ST428: design, transfer, robustness, and rank seven

8.1 Synthetic invariant design

From a pool of 1800 mean-zero unit vectors, pivoted QR selects 500 rows for the complete 365-coefficient sextic design. After column normalization, the selected design has condition number 14965.05, versus 18764.64 for a same-pool random design, an improvement factor 1.2539. This remains ill-conditioned and wholly synthetic. It is an algorithmic design result, not a detector or laboratory calibration.

8.2 Chernoff transfer sharpness

The rigorous adapted-cone result gives contraction at most 0.9800737084.... Bilinear Ulam discretizations on 21^2, 31^2, and 41^2 belief-pair grids give leading eigenvalues near 0.48202 and second-to-first modulus ratios

$$0.595845, qquad 0.595826, qquad 0.595751.$$

Their stability shows that the analytic cone bound is conservative. It does not provide an interval lower bound on the infinite-dimensional spectral gap, and the HMM fixture remains supplied rather than FIN-derived.

8.3 Nonlinear local forcing

At a localized minimum, the smallest tangent Hessian eigenvalue at the center is 86.33235.... On a radius-10^{-6} ball, the paid diagonal-Hessian variation is at most 21.90150..., leaving $\mu \geq 64.43086$. If a tangent perturbation has norm at most ϵ, then while the trajectory stays in this ball,

$$\|e(t)\| \leq e^{-\mu t}\|e(0)\| + \frac{\epsilon}{\mu}(1 - e^{-\mu t}).$$

For $\|e(0)\| < R/2$ and $\epsilon < \mu R/2 = 3.2215 \times 10^{-5}$, the ball is invariant. Time and forcing here are dimensionless supplied objects.

8.4 Why rank seven still does not inherit the global theorem

The rank-seven mediator shares the seven positive eigenvalues of the strict operator, including $\lambda_{\max} = 2.3421820411\ldots$, but its off-diagonal entries have 24 positive and 42 negative pairs. The positive-weight collision envelope used for the strict Laplacian therefore cannot be transferred. A sign split is valid but does not close the global gap. Equal spectra do not imply equal positivity geometry.

9 Falsification ledger and surviving interpretation

- **Global transition.** A nonzero lower endpoint is proved, but neither uniqueness nor the value of the transition is proved. The local crossing is explicitly not promoted.
- **IR boundary.** The singular attachment is now proved in one common tube. A full enlarged complement-sign topology cover remains absent.
- **Morse landscape.** Nine roots and indices are locally proved. The number nine is not globally exhaustive; heteroclinic edges are numerical.
- **Invariant algebra.** The odd/even degree-six modular census is new. The explicit degree-eight relation basis failed under the resource stop.
- **Dynamics.** A local forced-flow inequality is proved under a declared dimensionless flow. It does not derive a physical clock or bath.
- **Physical bridge.** No source for g, selector, dimensional calibration, preparation, instrument, environment, apparatus, or record was generated.

The deepest conclusion that survives this batch is mathematical rather than ontological: the supplied strict finite spectral object supports a rigorously controlled entropy–quadratic phase family, a compactifiable singular optimization branch, and a nontrivial invariant algebra. These structures are mutually compatible, but they do not yet form a theorem deriving experimentally testable physics from primordial information.

10 Recommended programs ST432–ST446

Program	Acceptance test	Priority	Expected status
ST432	Compute a sparse degree-eight modular kernel with durable checkpoints; lift across at least two primes and verify every retained relation directly over the integers. Reject a merely floating nullspace.	1	Hard exact algebra
ST433	Construct an interval branch-and-bound lower bound for V_g on the entire simplex over successive gain slabs inside $[2.2513440, 3.998]$. Each slab must identify all admissible global orbit types.	1	High-value proof
ST434	If ST433 isolates one crossing, certify equality of the competing branch values, transversality, and absence of a third competitor. This is the acceptance gate for a unique global critical gain.	1	Conditional on ST433
ST435	Cover a D_{12} fundamental domain of the eleven-simplex with interval Newton/Krawczyk boxes and exclusion boxes. Accept only if the nine-orbit stationary atlas is exhaustive.	1	Computationally hard

Program	Acceptance test	Priority	Expected status
ST436	Replace the ST422 integrations by isolating blocks or a Conley/Morse connection certificate for at least the uniform-to-localized index-one saddle.	2	Topological proof
ST437	Maximize the certified singular-attachment interval in b using adaptive parameter subdivision and prove where the common branch tube first fails or folds.	2	Likely feasible
ST438	Produce the missing complement-sign cover for the enlarged continuum-IR rectangle, including root ordering, derivative signs, inactive slack, and tail exclusion.	2	Proof-grade topology
ST439	Lift the joint odd/even generator census to characteristic zero and compute a minimal presentation through degree eight, including relations rather than only quotient counts.	2	Hard exact algebra
ST440	Optimize the sextic design for the smallest singular value under bounded observation noise; compare D-, E-, and minimax designs on a frozen candidate pool and hold-out pool.	3	Numerical design
ST441	Build a collectively compact finite-rank approximation of the synthetic Chernoff transfer with explicit discretization error, giving certified upper and lower spectral-gap brackets.	2	Functional analysis
ST442	Seek a global or cap-scale Lyapunov basin for the localized orbit under bounded tangent forcing, with positivity and simplex-boundary control.	3	Nonlinear dynamics
ST443	Formulate a sign-aware SDP or copositive relaxation for the rank-seven mediator and accept it only if the global value bound overlaps a certified feasible root.	3	Operator optimization
ST444	Reopen gain sourcing only if a genuinely new strict typed scalar source or normalization law appears; otherwise preserve ST429 without replay.	Gate	Source dependent
ST445	Reopen selector/orientation only if a new nonpremise strict provider escapes the exhausted D_{12}-invariant class; otherwise preserve QW-2191.	Gate	Provider dependent
ST446	Accept physical promotion only after independent preparation, calibrated clock, vertex-resolving instrument, raw event record, frozen hold-out, and separated custody are supplied.	Gate	External evidence

The recommended execution order is

$$\boxed{\text{ST433} \to \text{ST434}, \qquad \text{ST432} \to \text{ST439}, \qquad \text{ST437} \to \text{ST438}, \qquad \text{ST435} \to \text{ST436}.}$$

ST444–ST446 are event-driven gates, not invitations to repeat closed searches.

11 Reproducibility and provenance

The executable entry point is `fin_st417_st431_research.py`. It emits one aggregate JSON, fifteen program packets, one CSV ledger, and four figures. The independent test suite `test_fin_st417_st431.py` contains nineteen tests covering live matrix properties, the regularized face residual, stationary residuals, packet hashes, interval-status gates, numerical-label boundaries, invariant counts, and figures. All nineteen tests pass.

Principal internal inputs are the executable and result packets for ST342, ST378, ST391–ST395, and ST402–ST416, especially:

- `FIN_ST392_Primitive_Invariant_Generators_and_Syzygies.json`;
- `FIN_ST405_Uniform_Gain_Persistence_Tube.json`;
- `FIN_ST407_Certified_and_Sampled_Morse_Landscape.json`;
- `FIN_ST414_Expanded_IR_Krawczyk_Tube.json`;
- `FIN_ST415_Certified_Limiting_IR_Band_Loss_Face.json`.

The SHA-256 manifest supplied with this release covers the PDF, source, tests, aggregate results, ledger, program packets, and figures. No network source, laboratory datum, or external audit enters the computation.

12 Conclusion

This batch closes one previously explicit gap: the continuum-IR branch is now proved to attach to its compactified limiting face. It also turns the vague global gain-change statement into the finite enclosure $[2.251344001799089, 3.999]$ and upgrades the sampled stationary representatives to nine locally certified roots with certified Morse indices. The failures are equally informative. The degree-eight syzygy basis did not complete; the stationary atlas is not exhaustive; the rank-seven global envelope does not transfer; and no gain source, selector, physical scale, or empirical record has appeared. The correct next work is therefore focused: global branch-and-bound, checkpointed exact algebra, singular-topology completion, and only then any renewed physical interpretation.